AML Risk Prioritization

Model Development and Validation Report | Synthetic AML account-day prioritization

Portfolio objective: build a leakage-aware ranking model and convert it into an investigator-capacity policy that can be defended analytically, operationally, and through model-governance evidence.

Frozen Test Metric	Result
Population	494,707 account-days
Review rate	0.555%
Alerts prioritized	2,747
Positives identified	397
Positive yield	14.45%
Positive coverage	26.48%
Lift vs random	47.7x
ROC-AUC / PR-AUC	0.8529 / 0.1231

Executive takeaway

The final policy reviews only 0.555% of the in-distribution test population while concentrating positives about 47.7 times above random review. The project therefore treats the model as a ranking engine inside an operational review process, not as a stand-alone yes/no classifier.

1. Business framing

AML investigators have finite capacity, so the central question is not only whether an account-day is positive. The more useful decision is which account-days should be reviewed first and how queue size trades positive yield against positive coverage. This framing determines the evaluation metrics, policy-selection logic, and final Tableau design.

2. Data and analytical grain

Synthetic transaction data is transformed in MySQL into an account-centric view that represents both sender and receiver perspectives. Activity is aggregated to one row per account-day with transaction counts, direction ratios, sent/received amounts, amount statistics, bank/counterparty diversity, currency diversity, and rapid-transaction behavior. The Sep. 1-10 development extract contains 2,479,354 account-days and 7,131 positives (0.2876%).

3. Temporal validation and leakage control

Chronological windows are used rather than a random split: Sep. 1-6 train, Sep. 7-8 validation, Sep. 9-10 test. Sep. 11-18 is isolated for later OOD monitoring. Account ID and activity date are retained only for traceability and splitting, not as predictors. Model selection and review policy are completed before test evaluation.

4. Training-only EDA and preprocessing

Training EDA shows extreme right skew in counts and monetary features. The frozen preprocessing layer applies log1p to transaction count, sent/received amounts, average/max transaction amount, and unique-bank count. Ratios and currency diversity remain on their natural scale. The final model uses nine predictors.

5. Baseline and challenger

A class-balanced logistic regression provides the interpretable baseline (validation ROC-AUC 0.7995; PR-AUC 0.0378). At a default 0.50 threshold it produces more than 115,000 alerts, which is operationally unrealistic. XGBoost materially improves ranking quality (tracked validation PR-AUC about 0.13) and becomes the final challenger after targeted tuning.

6. Operating-policy selection

Validation scores are sorted and evaluated at fixed review capacities. The selected 2,500-alert validation policy balances queue size, positive yield, positive coverage, and lift. That review rate is frozen before test scoring, producing 2,747 test alerts because the test population is larger than validation.

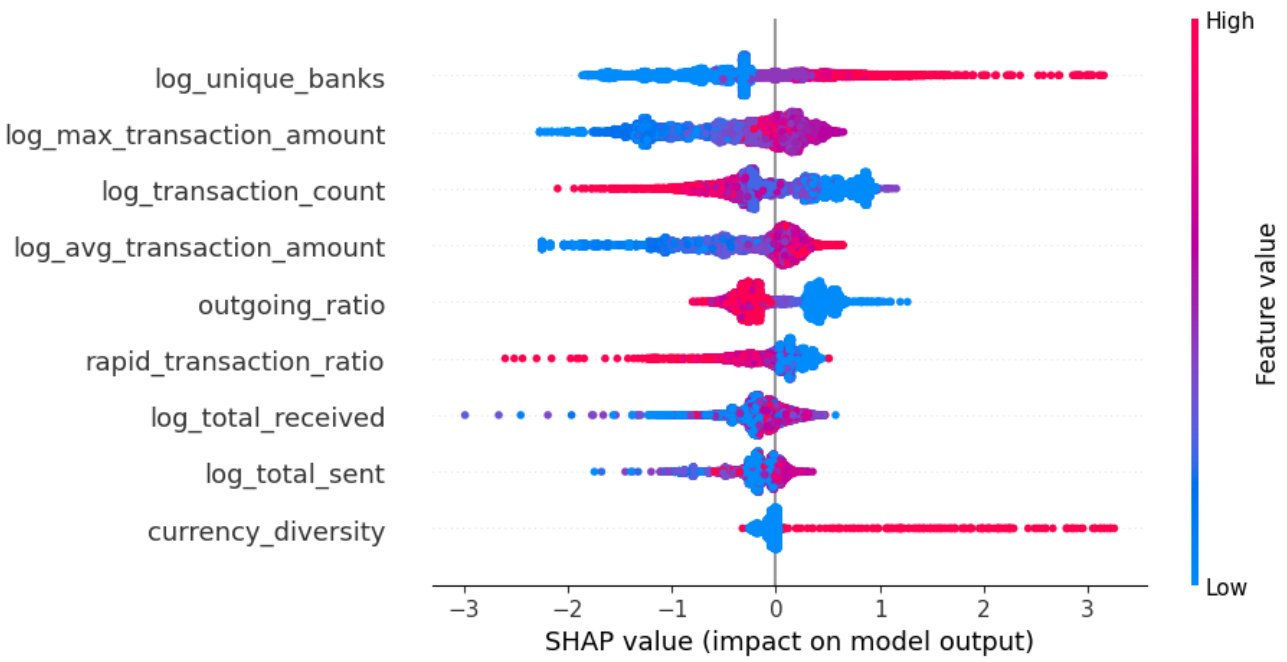
7. Final test results

Metric	Result
ROC-AUC	0.8529
PR-AUC	0.1231
Alerts reviewed	2,747
True positives	397
False positives	2,350
False negatives	1,102
Positive yield	14.45%

Positive coverage	26.48%
Lift vs random	47.7x
Test threshold	0.9034

8. Explainability

SHAP is used to explain both global feature importance and individual high-risk predictions. The results describe what the fitted model relies on; they are not treated as causal evidence.



Global SHAP beeswarm. High-value effects vary by feature; model interpretation is kept separate from causal interpretation.

9. Cost-sensitive policy analysis

A sensitivity analysis assigns one cost unit to an unnecessary review and varies the relative cost of a missed positive. The preferred tested capacity rises as missed-positive cost rises, demonstrating that queue size is a business-policy decision rather than a fixed property of model probability.

Missed-positive cost	Lowest-cost tested capacity
10x	1,000 alerts
25x	2,500 alerts
50x	10,000 alerts
100x	25,000 alerts
250x	50,000 alerts

10. Experiment tracking

MLflow logs model parameters, validation metrics, selected policy metrics, test metrics, and model artifacts. This preserves the logistic/XGBoost comparison outside transient notebook state and makes the development path auditable.

11. OOD monitoring and drift

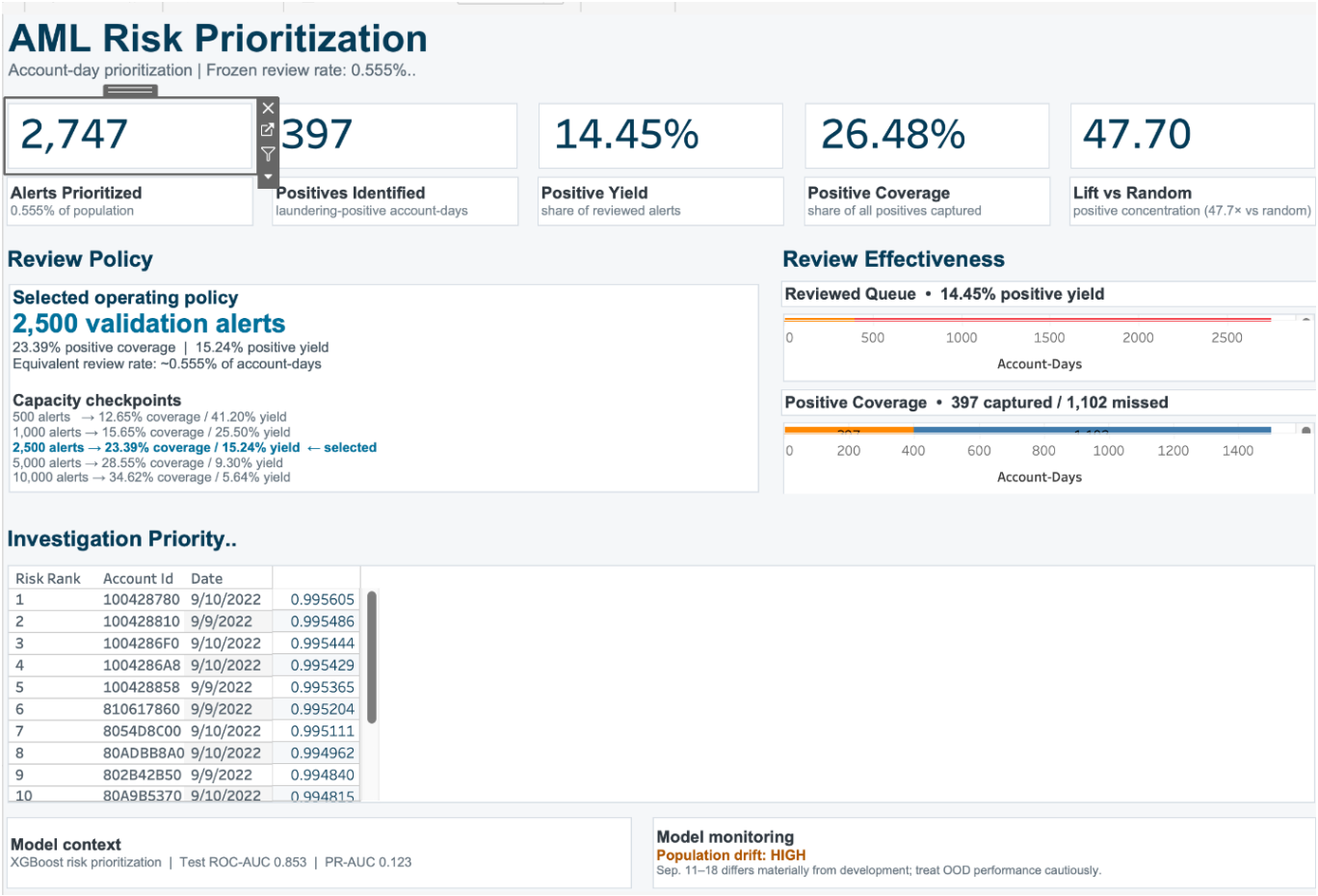
The Sep. 11-18 OOD extract contains only 923 account-days and 922 positives. PR-AUC of 0.9998 is therefore not evidence of extraordinary generalization because the baseline positive prevalence is already almost 1.0. The useful conclusion is distribution shift: robust PSI is severe for multiple transaction-amount, currency-diversity, and rapid-activity features. The

final dashboard labels population drift HIGH and cautions against ordinary performance interpretation.

Governance conclusion: model output remains usable as a stress-test signal, but the OOD sample does not provide a valid unbiased performance estimate.

12. Tableau decision-support artifact

The dashboard translates model metrics into operational language: alerts prioritized, positives identified, positive yield, positive coverage, lift versus random, review-policy checkpoints, review effectiveness, investigation priority, model context, and population-drift warning.



Final accepted Tableau portfolio dashboard. The packaged workbook is included in the repository.

13. Development hurdles and fixes

Hurdle	Fix / impact
Raw ingestion	Early Workbench imports produced truncation/schema issues; the workflow moved to an explicit schema and LOAD DATA LOCAL INFILE with reconciliation checks.
Class imbalance	A default probability threshold produced an impractical alert volume; top-K capacity analysis became part of evaluation.
MLflow result schema	Run names are stored under tags.mlflow.runName rather than a run_name column; the comparison query was corrected.
OOD interpretation	The later population is almost entirely positive, so performance metrics were reinterpreted as monitoring evidence rather than proof of stability.
Tableau rendering	Some generated workbook XML failed validation or rendered blank; the final workflow separates workbook logic QA from rendered screenshot QA and repairs only failed components.

14. Limitations

• Synthetic simulator data and labels do not reproduce real AML investigation complexity.

• Account-day aggregation simplifies customer/entity structure.

• The OOD sample is tiny and almost entirely positive.

• Cost ratios are sensitivity assumptions rather than institution-estimated losses.

• This is a prioritization prototype, not a production transaction-monitoring replacement.

15. Reproducibility and interview use

The accompanying SQL pipeline creates the account-day modeling views; the polished Colab notebook reproduces model development and export; the Tableau workbook provides the final decision-support artifact. The repository also includes a separate interview defense guide covering temporal validation, imbalance, lift, review capacity, SHAP, cost sensitivity, MLflow, OOD drift, and productionization.